

LAB 6

DHCP & RELAY

Centralized DHCP • ip helper-address Relay

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One router (R1) serves DHCP for two departments. HR is local; Sales sits behind a second router (R2) and is reached only because R2 relays DHCP broadcasts to R1 with ip helper-address.

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Overview & Topology

Two departments, two routers, one DHCP server. HR Department (VPC-1, VPC-2) sits directly on R1, so R1 can hand out addresses to it locally. Sales Department (Win, VPC-4) sits behind R2, a full router-hop away — DHCP is a broadcast protocol and broadcasts don't cross routers, so without help Sales would never see R1's DHCP offers. The fix is ip helper-address on R2: it turns incoming DHCP broadcasts from Sales into a unicast forwarded straight to R1, which answers using a second pool scoped to the Sales subnet.

- HR Department — VPC-1, VPC-2 — served by R1's local "HR" DHCP pool
- Sales Department — Win, VPC-4 — served by R1's second pool, reached only via R2's DHCP relay

IP Address Plan

Device	Interface	IP Address	Subnet Mask	Role
R1	e0/0	50.0.0.1	255.0.0.0	HR gateway
R1	e0/1	172.16.0.1	255.255.0.0	WAN → R2
R2	e0/1	172.16.0.2	255.255.0.0	WAN → R1
R2	e0/0	192.168.2.1	255.255.255.0	Sales gateway
VPC-1	eth0	50.0.0.51 (DHCP)	/8	HR client
VPC-2	eth0	50.0.0.52 (DHCP)	/8	HR client
Win	eth0	192.168.2.52 (DHCP)	/24	Sales client
VPC-4	eth0	192.168.2.53 (DHCP)	/24	Sales client

DHCP Pool Summary

Pool	Network	Default Router	DNS Server	Lease	Excluded Range
HR	50.0.0.0 / 255.0.0.0	50.0.0.1	8.8.8.8	123 days	50.0.0.1 – 50.0.0.50
Sales *	192.168.2.0 / 255.255.255.0	192.168.2.1	8.8.8.8	123 days	192.168.2.1 – 192.168.2.50

* Pool name scrolled off-screen in the captured terminal output — all other parameters confirmed directly from the CLI.

TASK 1 Topology & Interface Addressing (R1 & R2)

Both routers addressed and brought up before any routing or DHCP configuration.

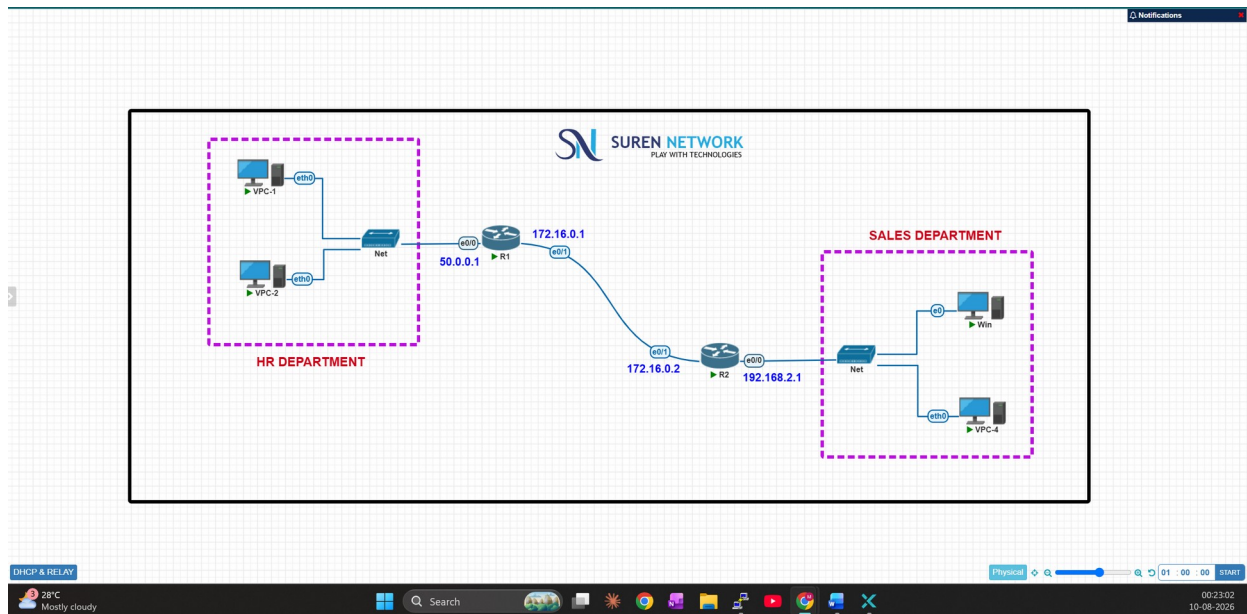


Figure 1 — Full topology: HR Department — R1 — R2 — Sales Department

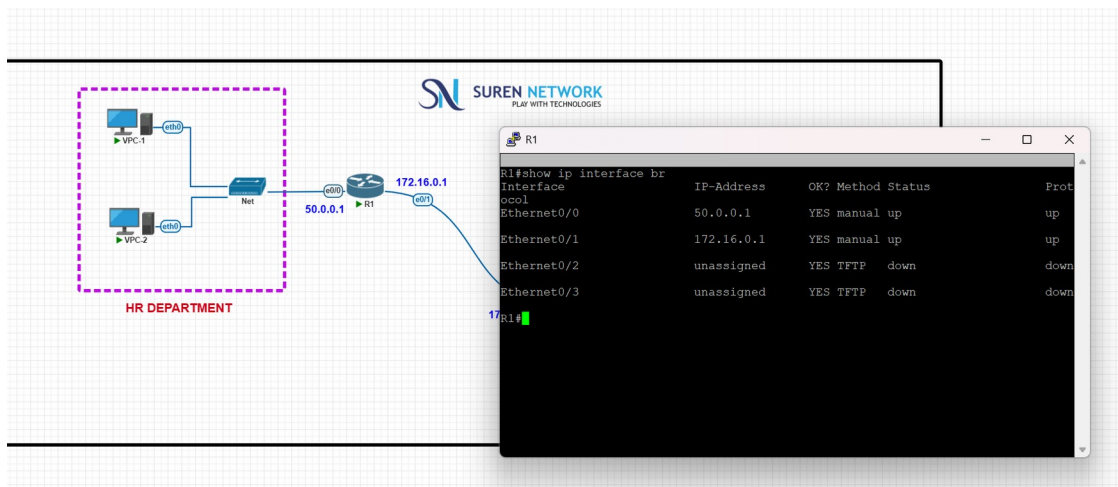


Figure 2 — R1: Ethernet0/0 = 50.0.0.1, Ethernet0/1 = 172.16.0.1 (show ip interface br)

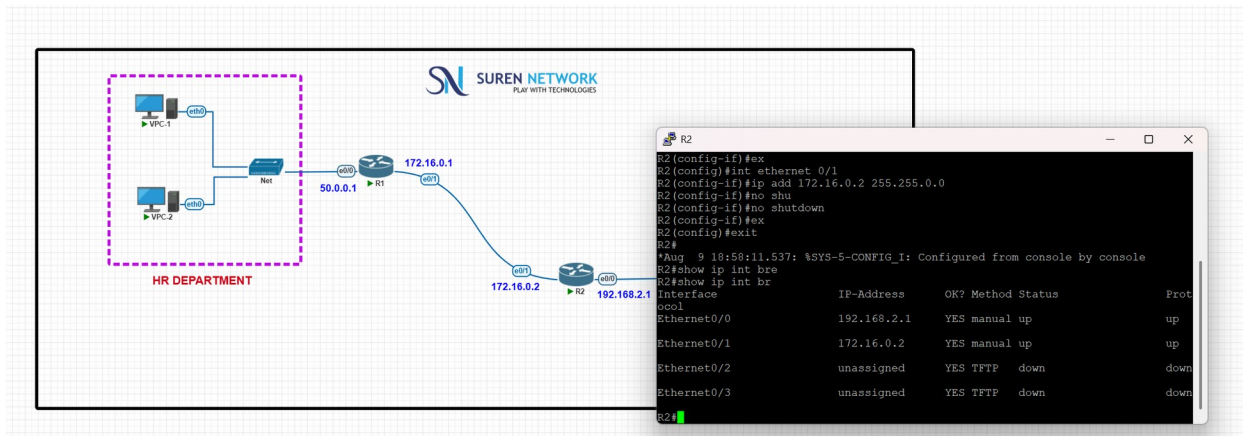
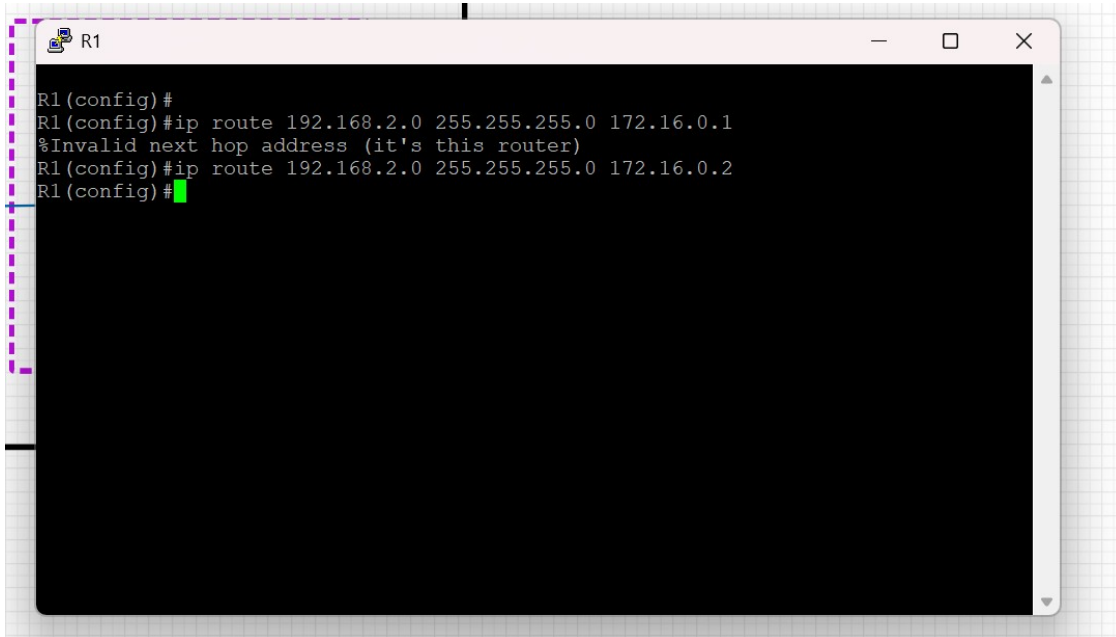


Figure 3 — R2: Ethernet0/1 = 172.16.0.2, Ethernet0/0 = 192.168.2.1

TASK 2 Inter-Router Static Routing

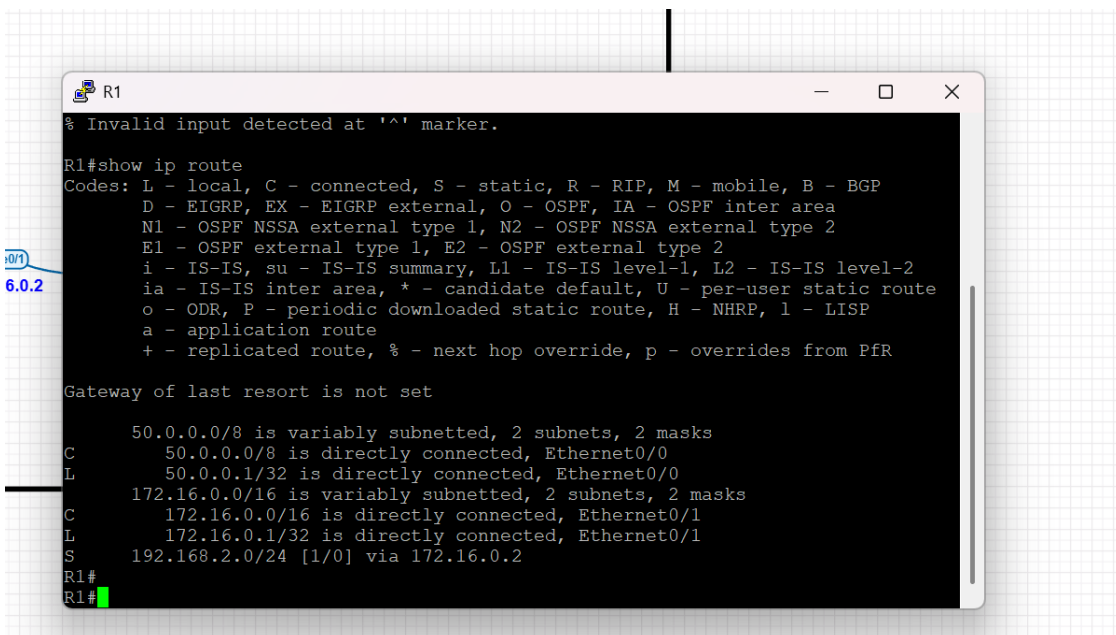
Each router needs a route to the other's LAN before DHCP relay (or any cross-network traffic) can work. First attempt on R1 used its own interface as the next hop and was correctly rejected.

```
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.1
%Invalid next hop address (it's this router)
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.2 ← corrected
```



```
R1
R1(config)#
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.1
%Invalid next hop address (it's this router)
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.2
R1(config)#
```

Figure 4 — R1: static route to Sales (192.168.2.0/24), first attempt rejected then corrected



```
R1
% Invalid input detected at '^' marker.

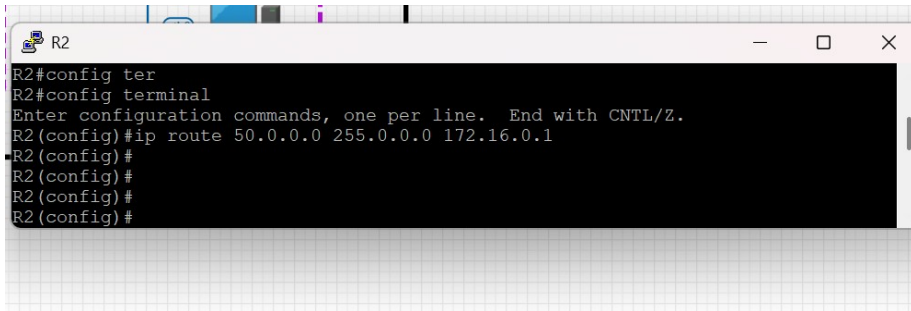
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

  50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       50.0.0.0/8 is directly connected, Ethernet0/0
L       50.0.0.1/32 is directly connected, Ethernet0/0
  172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Ethernet0/1
L       172.16.0.1/32 is directly connected, Ethernet0/1
S       192.168.2.0/24 [1/0] via 172.16.0.2
R1#
R1#
```

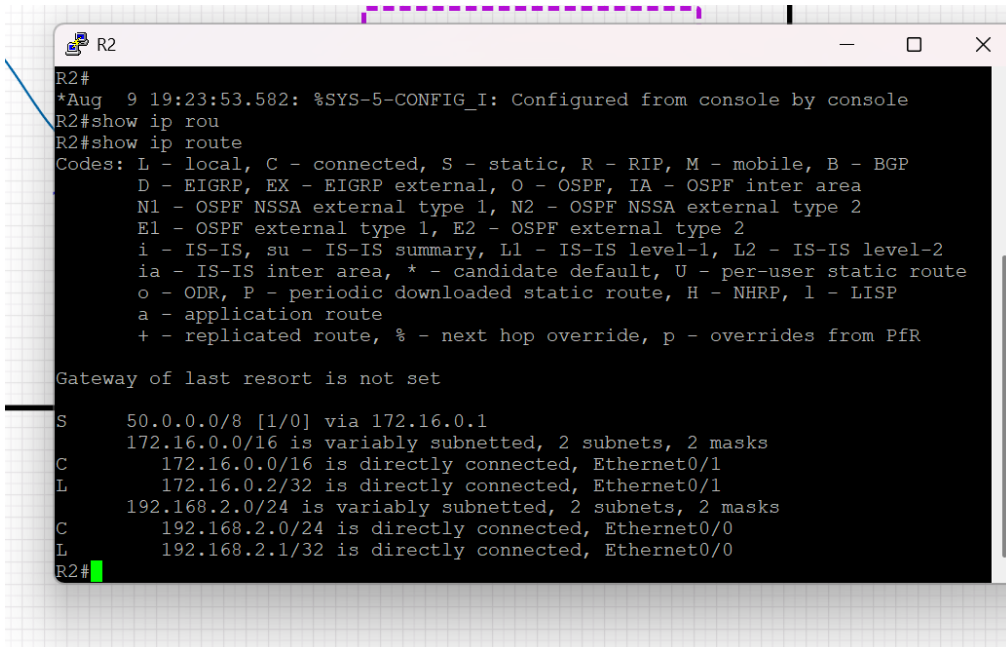
Figure 5 — R1: show ip route confirms the static route via 172.16.0.2

```
R2(config)#ip route 50.0.0.0 255.0.0.0 172.16.0.1
```



```
R2#
R2#config ter
R2#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 50.0.0.0 255.0.0.0 172.16.0.1
R2(config)#
R2(config)#
R2(config)#
R2(config)#
```

Figure 6 — R2: static route back to HR (50.0.0.0/8)



```
R2#
*Aug  9 19:23:53.582: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip rou
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

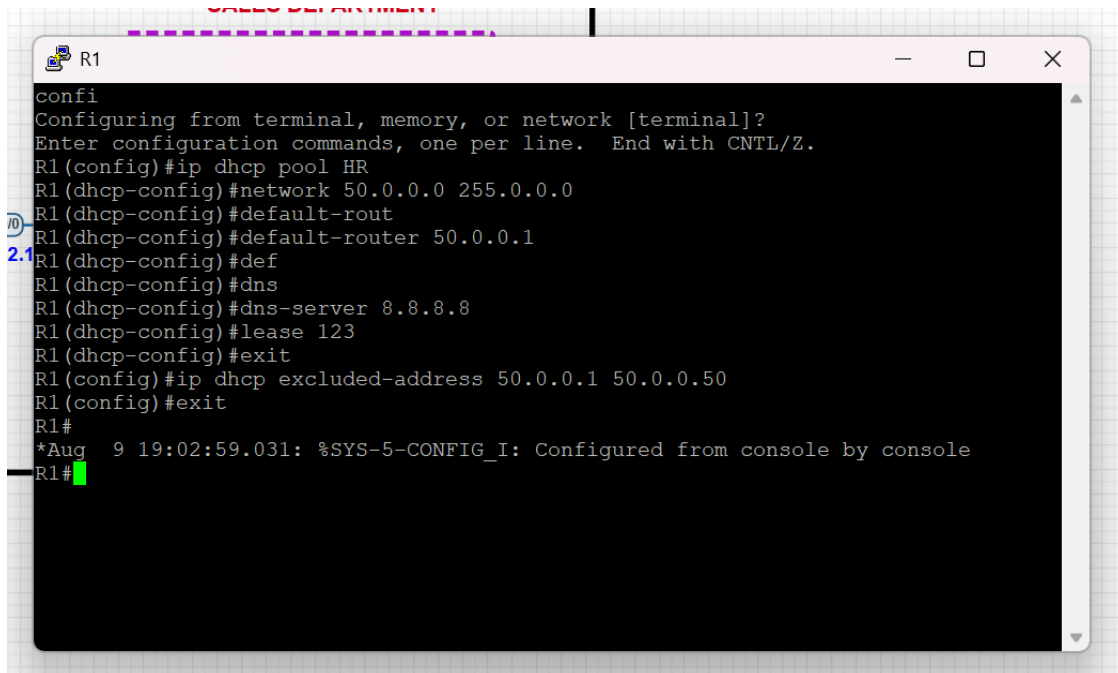
S    50.0.0.0/8 [1/0] via 172.16.0.1
C    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.16.0.0/16 is directly connected, Ethernet0/1
L      172.16.0.2/32 is directly connected, Ethernet0/1
L    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected, Ethernet0/0
L      192.168.2.1/32 is directly connected, Ethernet0/0
R2#
```

Figure 7 — R2: show ip route confirms both connected and static routes

TASK 3 Configuring the DHCP Pool for HR Department

DHCP pool created directly on R1 for the locally-attached HR subnet.

```
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 50.0.0.0 255.0.0.0
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 123
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50
```



```
R1
confi
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 50.0.0.0 255.0.0.0
R1(dhcp-config)#default-rout
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#def
R1(dhcp-config)#dns
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 123
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50
R1(config)#exit
R1#
*Aug 9 19:02:59.031: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Figure 8 — R1: HR DHCP pool — network, default-router, dns-server, lease, excluded range

lease 123 sets a 123-day lease (days only — no hours/minutes given). Leases issued Aug 9 2026 expiring “Dec 10 2026” in Task 4's binding table line up exactly with that math.

TASK 4 Verifying DHCP on HR Clients (VPC-1, VPC-2)

Both HR clients pull an address with a simple ip dhcp request, landing right after the excluded range as expected.

- VPC-1 → 50.0.0.51/8, gateway 50.0.0.1
- VPC-2 → 50.0.0.52/8, gateway 50.0.0.1

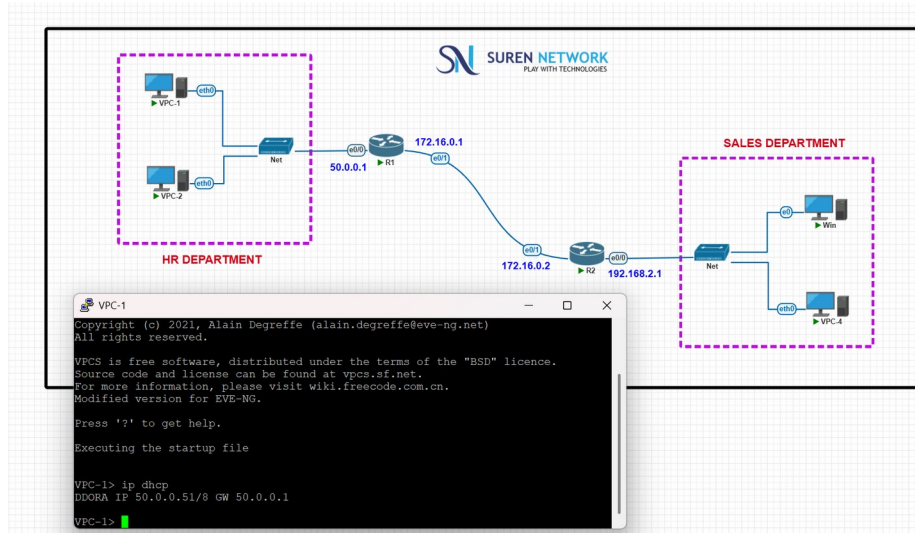


Figure 9 — VPC-1 receives 50.0.0.51 via DHCP (DDORA)

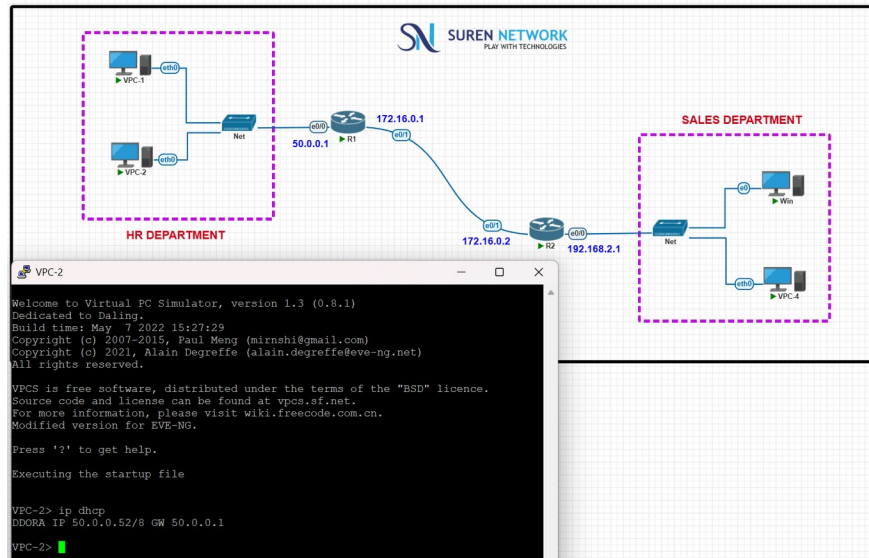


Figure 10 — VPC-2 receives 50.0.0.52 via DHCP (DDORA)

R1's binding table confirms both leases, each with its client MAC and lease expiry.

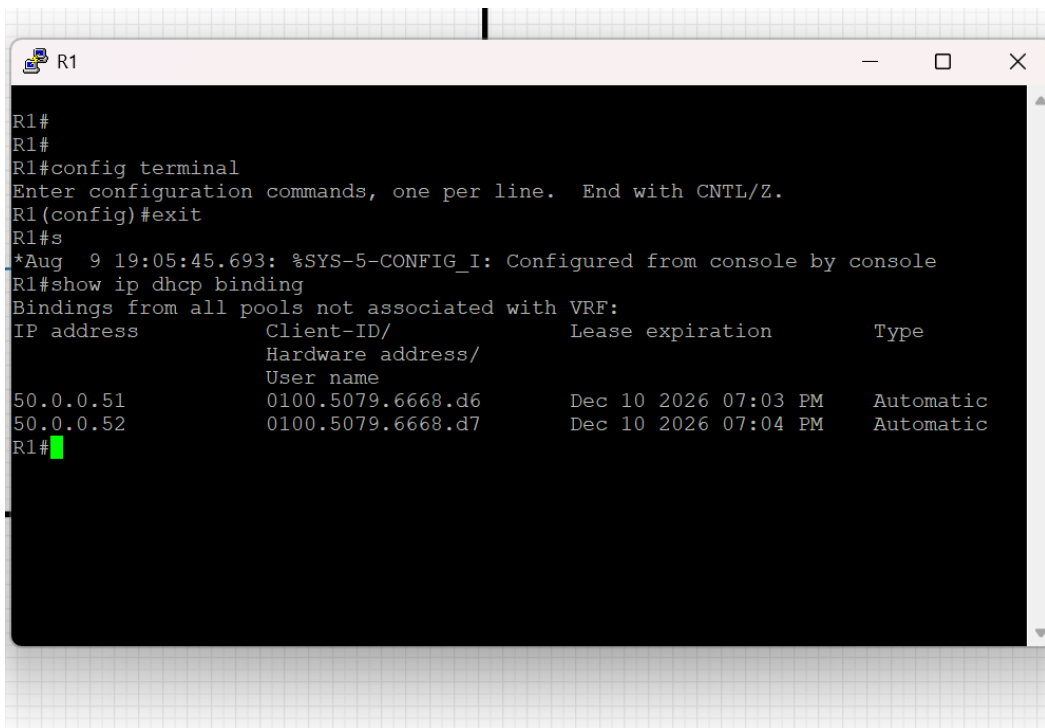


Figure 11 — R1: show ip dhcp binding — both HR leases recorded

TASK 5 Configuring the DHCP Pool for Sales Department

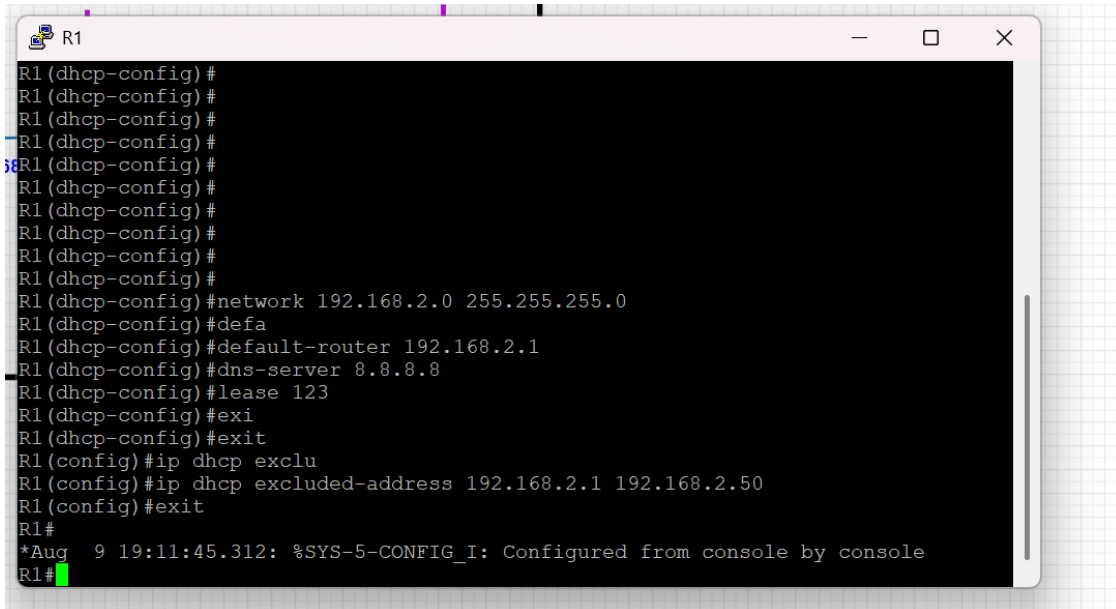
A second pool on R1, scoped to the Sales subnet even though R1 has no interface on it directly — this is what the relay in Task 6 will deliver requests to.

```

R1(dhcp-config)#network 192.168.2.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.2.1
R1(dhcp-config)#dns-server 8.8.8.8

```

```
R1(dhcp-config)#lease 123
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.50
```



```
R1
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#
R1(dhcp-config)#network 192.168.2.0 255.255.255.0
R1(dhcp-config)#defa
R1(dhcp-config)#default-router 192.168.2.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 123
R1(dhcp-config)#exi
R1(dhcp-config)#exit
R1(config)#ip dhcp exclu
R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.50
R1(config)#exit
R1#
*Aug 9 19:11:45.312: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Figure 12 — R1: second DHCP pool for the Sales subnet (192.168.2.0/24)

TASK 6 Configuring DHCP Relay (ip helper-address) on R2

R2's Sales-facing interface is told where to forward DHCP broadcasts it receives — straight to R1, the DHCP server.

```
R2(config)#interface ethernet 0/0
R2(config-if)#ip helper-address 172.16.0.1
```

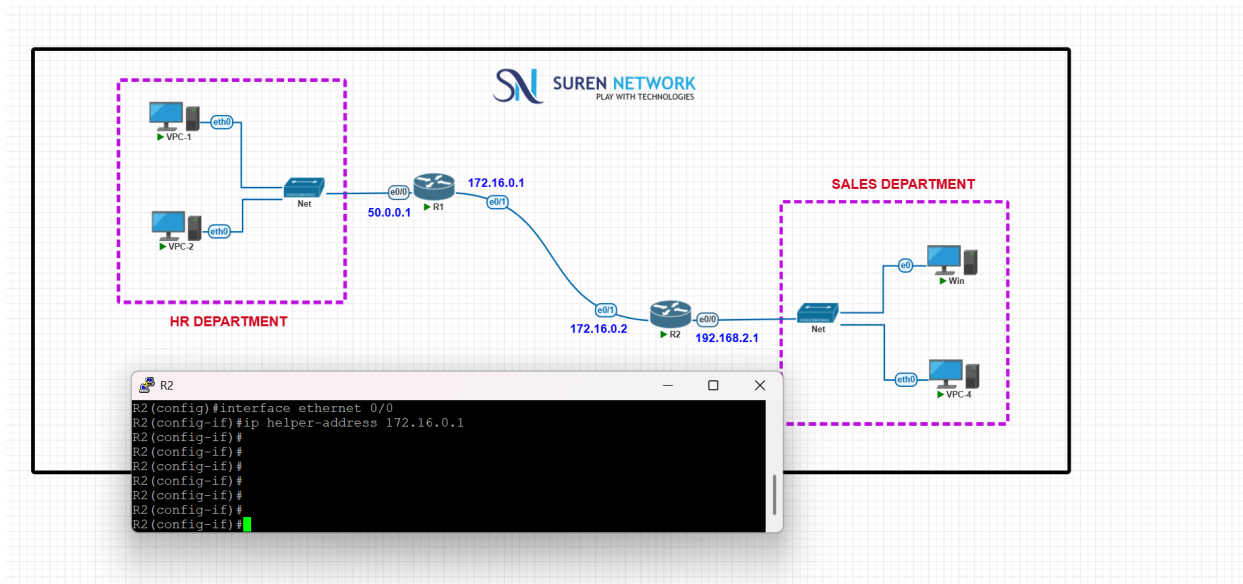


Figure 13 — R2: ip helper-address 172.16.0.1 configured on Ethernet0/0

TASK 7 Verifying DHCP Relay on Sales Clients (Win, VPC-4)

With the relay in place, Sales clients now reach R1's second pool despite being a router-hop away — proof the relay is working end-to-end.

- Win (ipconfig) → 192.168.2.52, mask 255.255.255.0, gateway 192.168.2.1
- VPC-4 (ip dhcp) → 192.168.2.53/24, gateway 192.168.2.1

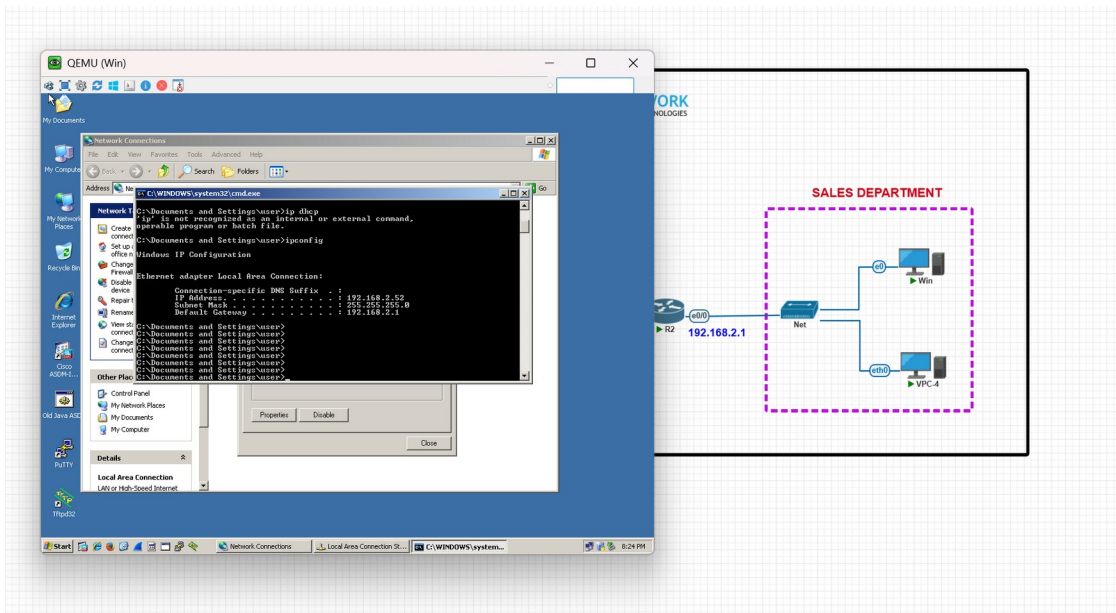


Figure 14 — Win PC receives 192.168.2.52 via relayed DHCP

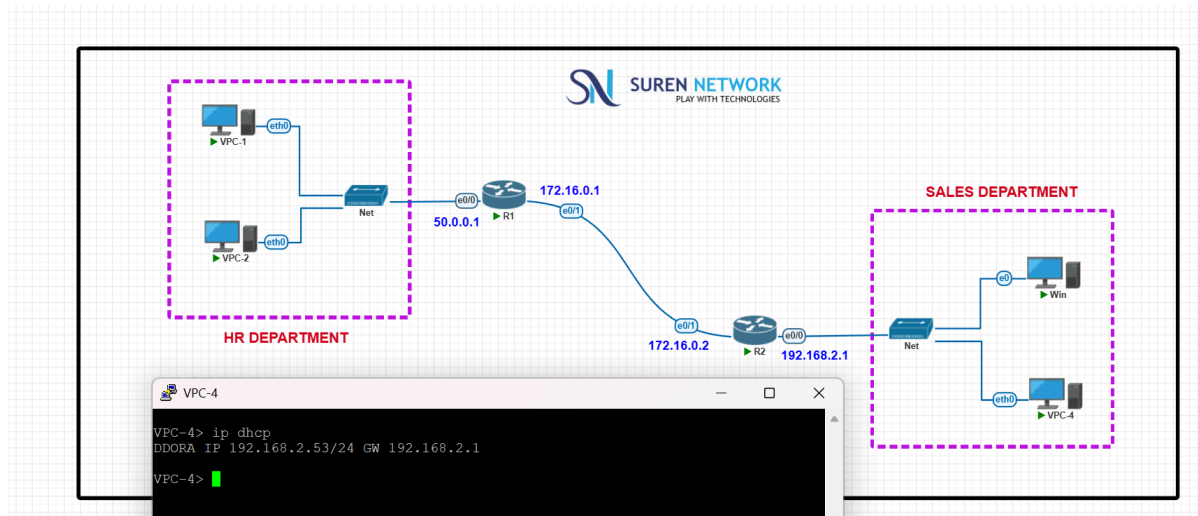


Figure 15 — VPC-4 receives 192.168.2.53 via relayed DHCP (DDORA)

Conclusion

R1 successfully serves DHCP for two separate subnets from two pools: HR directly, and Sales entirely through R2's ip helper-address relay. Both HR clients (VPC-1, VPC-2) and both Sales clients (Win, VPC-4) obtained correct addresses, masks, gateways, and DNS settings with no manual configuration on the end devices — confirming that static routing between R1 and R2, the two DHCP pools, and the relay configuration are all working together correctly.